

Rahul Guruprakash

rahulguruprakash@gmail.com | +1 (858) 305-6369 | [g-rahul.github.io](https://github.com/rahul-g) | linkedin.com/in/rahul-g-055720175

Education

North Carolina State University

Master of Science in Computer Engineering

Raleigh, NC

Expected May 2027 | GPA: 4.0/4.0

R.V. College of Engineering

Bachelor of Engineering in Electrical and Electronics Engineering

Bangalore, India

August 2021 | GPA: 8.3/10

Technical Skills

Programming: C, C++, Python, CUDA, OpenMP, Arm assembly (AArch64), Bash, Perl.

Systems: Linux kernel/drivers, Zephyr, UEFI/EDK II, ACPI, Device Tree, FF-A, TF-A, Fixed Virtual Platforms, QEMU.

Build, Test & Debug: Yocto/OpenEmbedded, Buildroot, CMake, SCons, GCC, Clang/LLVM, GoogleTest, Trace32, GDB.

Professional Experience

Arm – Intern, Silicon Engineering Team, Austin, TX

May 2026 – Present

- Defined the system-level validation strategy for the CMN interconnect and AI accelerator; developed kernel and user-space tests on Fixed Virtual Platforms and brought them up on silicon. Added system-level memory testing for DDR/HBM traffic to emulate AI workloads and system behavior across firmware, the kernel, and ML runtime services.
- Brought up Yocto/OpenEmbedded extensions for platform firmware and created a customized Linux root filesystem for SoC QA.

Arm – Engineer, System Compliance (ATG), Bangalore, India

July 2023 – August 2025

- Delivered Arm's open-source [FF-A Architecture Compliance Suite \(ACS\)](#) releases for FF-A v1.1 and v1.2, adding self-checking tests for secure-partition messaging, memory sharing, interrupts, and notifications.
- Identified FF-A conformance issues across TF-A, Hafnium, and Xen and helped drive fixes through ACS testing.
- Expanded bare-metal FF-A coverage on Arm Base FVP using MMU/GIC support and an SMMUv3 test engine, covering memory permissions, interrupt delivery, and DMA isolation.
- Designed a Linux kernel driver and user-space port for FF-A ACS, enabling FF-A compliance validation from Linux.

Qualcomm – Engineer, RF Software, Hyderabad, India

October 2021 – July 2023

- Developed RF-control firmware for the Snapdragon X35 5G Modem-RF System across the modem DSP and a dedicated RF-control processor, supporting wireless-transceiver image loading, initialization, and RF-front-end programming.
- Built GoogleTest-based unit tests for firmware validation, catching functional and integration bugs during virtual-platform testing, FPGA emulation, and silicon bring-up.
- Added software CRC checks to RF-control firmware to verify the integrity of DMA payloads received from the modem DSP.
- Mitigated an instruction-decoder hardware issue through a firmware image-layout workaround and coordinated toolchain support with the compiler team.

Log9 Materials – Engineer, Embedded Systems, Bangalore, India

January 2021 – September 2021

- Implemented a GB/T 27930 EV charging protocol stack on an Arm Cortex-M MCU, including SAE J1939/21 message handling and multi-packet transport over CAN.
- Stabilized firmware for production on 100+ electric three-wheelers; built a CAN bootloader, test framework, and validation jig.

Academic Research & Projects

Graduate Research Assistant (Part-time) – North Carolina State University, Raleigh, NC

August 2025 – May 2026

- Independently designed both the hardware architecture and high-speed PCB layout for a custom NVIDIA Jetson Orin NX carrier board used in an embedded AI biomass-mapping system, including PCIe Gen4, Gigabit Ethernet, and USB 3.2 interfaces.

Computer Architecture Projects – North Carolina State University

Spring 2026

- **Advanced Microarchitecture (ECE 721):** Co-developed a 30 KB hybrid value predictor for an out-of-order RISC-V simulator; the team design improved harmonic-mean IPC by 1.24% over a stride-predictor baseline across 15 SPEC CPU benchmarks.
- **GPU Architecture (ECE 786):** Extended Accel-Sim with a per-SM, per-kernel profile-guided L1 data-cache bypass policy, improving simulated IPC by 2.5–42.8% over no bypass across four cache-unfriendly workloads.